SAKTHIVEL S

MECHANICAL ENGINEERING | CAD DESIGN | PRODUCT DEVELOPMENT

+91 90 800 39 122 | sakthiveldesigner05@gmail.com | Hosur, Tamil Nadu, India
Portfolio: https://sakthiveldesigner.vercel.app/

PROFESSIONAL SUMMARY

Mechanical Engineering undergraduate specializing in CAD modelling, product design, and manufacturing engineering. Proficient in CATIA V5, SolidWorks, AutoCAD, and 3DEXPERIENCE for design and engineering analysis. Experienced in multidisciplinary projects involving robotic welding, electric mobility, hydrogen production, and AI applications. AICTE Innovation Student Ambassador and Best Research Paper Award recipient, passionate about innovative engineering solutions.

EDUCATION

B.E. Mechanical Engineering | Anna University | 2027
Er. Perumal Manimekalai College of Engineering, Hosur | CGPA: 7.5

HSC (XII) | Tamil Nadu State Board | 2023
Govt. Hr. Sec. School, Panjappalli, Dharmapuri | 61%

SSLC (X) | Tamil Nadu State Board | 2021
Govt. Hr. School, D. Rathinagiri, Dhadikal, Krishnagiri | All Pass

TECHNICAL SKILLS

CAD CATIA V5 (Part Design, Generative Shape Design, Assembly Design, Drafting), SolidWorks, AutoCAD, 3DEXPERIENCE.

Engineering Engineering drawings, GD&T fundamentals, component design, material selection, DFM/DFA, basic FEA.

Manufacturing Machining, welding, casting, forging, automation fundamentals.

Digital MS Excel, PowerPoint, technical documentation; exposure to AI workflow automation and computer vision.

SELECTED ACADEMIC PROJECTS

Automated Multi-Process Welding Robot | Design Concept
Designed a system concept with interchangeable end-effectors; explored arm design, workspace planning, tool changing, and motion automation.

Self-Charging Electric Vehicle Concept | Electric Mobility
Studied auxiliary energy recovery, battery monitoring, energy management, and regenerative braking.

Hydrogen Production System | SIH 2025
Explored water electrolysis, renewable energy integration, process automation, and hydrogen storage concepts.

CAD-Based Mechanical Design & Simulation | Mechanical Design
Created component models and engineering drawings using CATIA V5, SolidWorks, and AutoCAD; focused on assembly fitment and manufacturability.

AI Application Concepts | Workflow Automation & Computer Vision
Designed concepts for an agentic business assistant and CCTV-based smart attendance system using workflow automation and computer vision.

EDUCATION

B.E. Mechanical Engineering | Anna University | Expected 2027
Er. Perumal Manimekalai College of Engineering, Hosur | CGPA: 7.5/10

HSC | 2023
Government Higher Secondary School, Panjappalli, Dharmapuri | 61%

SSLC | 2021
Government High School, D. Rathinagiri, Dhadikal, Krishnagiri | Passed

INTERNSHIP & INDUSTRIAL TRAINING

Virya Industries Pvt. Ltd. | Hosur | Dec 2025–Jan 2026
Exposure to industrial operations, production planning, workflow coordination, documentation, and process improvement.

Adroid Automation India Pvt. Ltd. | Hosur | Mar 2024
Exposure to industrial automation, equipment operation, manufacturing workflows, and technical reporting.

CERTIFICATIONS

- CATIA V5 and 3DEXPERIENCE Certifications - Dassault Systèmes / TANCAM.
- AICTE Innovation Student Ambassador.
- Best Research Paper Award - Annamalai University.

ACHIEVEMENTS

- EDP Student Innovation Ambassador: Supported entrepreneurship activities, student pitching, and an eight-hour startup challenge.
- Innovation Student Coordinator: Supported innovation and intellectual property awareness.

SOFT SKILLS

Problem-solving; analytical thinking; product development; innovation; teamwork; leadership; project coordination; technical communication; continuous learning; Vibe Coding.